

Neuromorphic Hyperdimensional Computing on Edge MEMS for Sub-Microwatt Cardiac Anomaly Detection

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1. Introduction

Continuous wearable cardiac telemetry requires real-time electrocardiogram (ECG) arrhythmia classification on ultra-low-power biomedical edge devices. Conventional deep neural networks incur prohibitive memory footprints and memory bandwidth bottlenecks, exceeding the micro-watt energy budgets of miniaturized implantable sensors. This research introduces a spike-driven Hyperdimensional Computing (HDC) architecture that maps continuous multi-channel physiological signals into pseudo-orthogonal 8,192-bit holographic vector representations.

2. Methodology

Our proposed framework couples an asynchronous event-based delta modulator with a hardware-friendly ternary hyperdimensional encoder. Raw voltage samples from the MIT-BIH Arrhythmia Database are converted into sparse temporal spike trains. A randomized continuous item memory projects QRS complex intervals into high-dimensional hyperspace using non-linear circular convolution and permutation operators. Single-pass online associative learning is performed directly on-chip using localized hardware XNOR-popcount primitives, eliminating off-chip gradient backpropagation overheads entirely.

3. Results

Evaluated against 48 patient records from PhysioNet MIT-BIH benchmarks, the neuromorphic HDC pipeline achieves a 97.4% ventricular ectopic beat classification accuracy and 98.1% F1-score. Synthesized in a 22nm FD-SOI semiconductor process, the full inference core consumes only 840 nanowatts at 0.5V supply voltage, delivering a 24x energy efficiency advantage compared to quantized edge CNN baselines.

4. Conclusion

This work validates spike-driven hyperdimensional vector representations as an ultra-low-power computing paradigm for continuous biomedical diagnostics. Future iterations will fabricate monolithic silicon prototypes with integrated piezoelectric energy harvesting.